Nathan Sujatno

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EDUCATION

The University of Texas at Dallas, Richardson, TX — May 2027
Bachelor of Science, Computer Science — GPA: 3.87

- Honors & Awards: Academic Excellence Scholar, Hobson Wildenthal Honors College

SKILLS

Languages: Python (Asyncio, Pydantic), C++, JavaScript/TypeScript, SQL **AI & Agents:** LangChain, RAG, Vector Databases, LLM Tool Use, Prompt Engineering (Chain-of-Thought), Embeddings **Backend & Data:** FastAPI, Node.js, MongoDB, Crawl4AI (Automated Scraping), REST APIs **Frontend & Tools:** React, Tailwind CSS, AWS, Git, Postman

EXPERIENCE

NRVE intern, Remote — June 2025 – August 2025
Backend Project Lead, Frontend Developer

- Architected and deployed a serverless backend on **AWS** (Lambda, API Gateway, S3) and **MongoDB**, building scalable **RESTful API** endpoints in Python.
- Cut database query latency by over **95%** by implementing a caching solution with AWS **CloudFront**.
- Contributed to frontend UI development and implemented the full integration layer between backend services and the mobile application.

PROJECTS

SWE Job Matcher — November 2025

- Architected a multi-step autonomous agent utilizing **LLM Tool Use** to orchestrate asynchronous web scraping and resume analysis in parallel.
- Engineered a deterministic evaluation framework using **Chain-of-Thought** prompting and **Pydantic/JSON parsing** to constrain LLM outputs into strict scoring metrics.
- Implemented a concurrent execution pipeline, optimizing the agent's token throughput and **reducing multi-step inference latency by 80%.**

Kanban Sync, HackUTD — November 2025

- Engineered a multi-source retrieval system that routes queries to distinct vector spaced for **dynamic operational state** and **static technical knowledge**, preventing hallucinations on rapidly changing data.
- Developed a **self-correcting validation loop** where the LLM autonomously cross-references user requests against retrieved datacenter state, rejecting invalid actions before execution.
- Implemented **JSON-enforced Chain-of-Thought** workflows to classify ticket urgency, combining heuristic metadata with semantic analysis to automate incident prioritization.

MeteorMate — June 2025 – Current

- Developed a responsive user interface with **Next.js**, creating a library of reusable components and leveraging **Tailwind CSS** for rapid, utility-first styling and consistent user experience.
- Architected a serverless backend using Firebase for authentication and Supabase, providing secure and scalable **RESTful** endpoints.

LEADERSHIP & ORGANIZATIONS

AI MD — September 2025 – December 2025
Project Manager

- Leading a student team in developing a full-stack MERN app that utilizes a custom-trained AI model.
- Managed project timelines, delegated tasks, and mentored team members in a collaborative environment.

ACM, Officer — May 2025 – Current
ACM TIP (Technical Interview Prep) — September 2025 – December 2025
ACM, Member — February 2025 – May 2025